Reg. No. | | | | | | | | | | | [14EC65]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

POWER ELECTRONICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by selecting at least TWO from each part
2. Missing data may be suitably assumed.

PART - A

1. a. Explain the control characteristics of MOSFET, MCT and SITH. **09**

   b. Discuss the peripheral effect during switching of power devices and also mention the remedies. **05**

2. a. With the help of the circuit diagram, explain the operation of Base Drive Control during turnoff. **07**

   b. With the necessary waveforms, explain the switching characteristic of MOSFET. **07**

3. a. Draw the circuit diagram, waveforms of a UJT Triggering circuit & explain it. **07**

   b. The Latching current of an SCR used in a phase controller circuit comprising an Inductive load of R=10Ω and L=0.1H is 15 mA. The input voltage is 325 sin314t. Obtain the minimum gate pulse width required for reliable triggering of the SCR if gated at π/3 angle in every +ve half cycle. **07**

4. a. Explain the operation of a 1-Φ semi converter with inductive load. **07**

   b. The 1- Φ dual converter is operated from 120 V, 60 Hz supply with load resistance of R=10 Ω.The circulating inductance is Lc = 40 mH, delay angles are α_1= 60° and α_2=120°. **07**

   Calculate the Peak circulating current and Peak current of converter-1

PART - B

5. a. Explain the operation of a 1-Φ AC voltage controller with inductive load. **07**

   b. A voltage source V_s=100 sin 377t supplies a resistive load of 100Ω through a pair of back connected thyristors. Calculate **07**

   i. $V_{o(rms)}$

   ii. Average power in the load

   Assuming that both thyristors fires at 45 degrees with respect to the supply voltage.

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6. a. Explain the self-communication with the help of sketch, waveform and obtain the expression for the capacitor voltage and current. **10**

b. Compare Natural and Force Commutation. **04**

7. a. Explain the principles of operation of a step up chopper with suitable circuit diagram and waveforms **07**

b. A DC chopper has a resistive load of 20 Ω and input voltage of 220 V. When chopper is on, its Voltage drop is 1.5 V and chopping frequency is 10khz. If duty cycle is 80 %, determine **07**

i. Average output voltage

ii. RMS value of output voltage

iii. Chopper on time

8. a. Explain the following performance parameters of inverters **06**

i. Harmonic factor of nth harmonic

ii. Total harmonic distortion

iii. Distortion factor

b. Draw the circuit diagram of a variable DC link inverter and explain it. **08**